Low-density parity-check codes

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A linear code is described by a parity-check matrix. When it has few ones, each check involves only a few positions of the word.

Sparse parity checks

Definition 0.1 (Regular low-density parity-check code)**.** Let $j, k \geq 1$ be integers. A (j, k)**-regular LDPC code** (low-density parity-check code) consists of the words $y \in \mathbb{F}_2^n$ with $Hy = 0$, where H has j ones in every column and k ones in every row, and zeros elsewhere. The rows of H need not be independent. When no row is a sum of other rows, H is a parity-check matrix of this linear code.

Example 0.1 (A $(2, 3)$-regular code of length 6)**.**
Take four items, labeled $1, 2, 3, 4$, and the six pairs of them. Let H have a row for each item and a column for each pair, with a 1 where the item lies in the pair. Every pair holds two items and every item lies in three pairs, so H has two ones in each column and three in each row, and it gives a $(2, 3)$-regular LDPC code of length 6. The R chunk below builds H, checks the counts, and counts the words y with $Hy = 0$ modulo 2.

```
pairs <- t(combn(4, 2))
H <- sapply(seq_len(nrow(pairs)), function(e) as.integer(1:4 %in% pairs[e, ]))
H
#>      [,1] [,2] [,3] [,4] [,5] [,6]
#> [1,]    1    1    1    0    0    0
#> [2,]    1    0    0    1    1    0
#> [3,]    0    1    0    1    0    1
#> [4,]    0    0    1    0    1    1
colSums(H)
#> [1] 2 2 2 2 2 2
rowSums(H)
#> [1] 3 3 3 3
words <- as.matrix(expand.grid(rep(list(0:1), 6)))
is_codeword <- apply(words, 1, function(y) all((H %*% y) %% 2 == 0))
sum(is_codeword)
#> [1] 8
```

There are 8 codewords, so the dimension is $\log_2$ of that number, which is 3.

Theorem 0.1 (The rate of a regular LDPC code)**.** *Let C be a (j, k)-regular LDPC code of length n. Then the rate of C is at least $1 - j/k$.*

Proof

Proof. Let H be as in the definition of C, with m rows and n columns. Count the ones of H in two ways. Each of the n columns holds j ones, so H has nj ones. Each of the m rows holds k ones, so H has mk ones. So $nj = mk$, and $m = nj/k$.
The codewords are the words y with $Hy = 0$, so their number is 2^{n-r}, where r is the rank of H. The rank of H is at most its number of rows, so $r \leq m$, and C has at least 2^{n-m} codewords. Its rate is $(n - r)/n$, which is at least $(n - m)/n = 1 - j/k$. □

Example 0.2 (The rate of the length-6 code)**.**
For the $(2, 3)$-regular LDPC code of Example 0.1, Theorem 0.1 gives a rate of at least $1 - 2/3$. It has 8 codewords, so its rate is $\log_2(8)/6$.

```
1 - 2 / 3
#> [1] 0.3333333
log2(8) / 6
#> [1] 0.5
```

The rate is 1/2, which is larger than 1/3. The four rows of H add to zero modulo 2, because every column has two ones, so H has rank 3, not 4.

Remark 0.1 (Where LDPC codes come from). Gallager (1962) introduced low-density parity-check codes, and MacKay (2003) covers them in his book.

References

Gallager, Robert G. 1962. "Low-Density Parity-Check Codes." *IEEE Transactions on Information Theory* 8 (1): 21–28. https://doi.org/10.1109/TIT.1962.1057683.

MacKay, David J. C. 2003. *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press.